

# Pneumonia Prediction Utility

By: Tyler Haisman

## Overview

My pneumonia prediction utility utilizes deep learning models to analyze chest X-ray images and predict whether pneumonia is present and, if so, whether it is bacterial or viral in nature.

## Technology Stack

### Frontend:

- Framework: Next.js (React)
- Styling: Tailwind CSS
- Language: TypeScript

### Backend:

- Framework: Django (Python)
- Deep Learning Library: TensorFlow Keras

## Models

### 1. Chest X-ray Detection Model:

- This model determines whether the image is a chest X-ray or not.
- Trained on nearly 2000 chest X-ray images.
- Utilizes a convolutional neural network (CNN) architecture.

### 2. Pneumonia Classification Model:

- This model identifies if pneumonia is present and categorizes it as bacterial or viral.
- Trained on over 4000 chest X-ray images depicting normal, bacterial pneumonia, and viral pneumonia cases.
- Utilizes a CNN architecture.

## Data Sources

### Chest X-Ray Images (Pneumonia) Dataset:

- Kaggle Dataset: Chest X-Ray Images (Pneumonia)
- Contains chest X-ray images depicting normal, bacterial pneumonia, and viral pneumonia cases.
- Kermany, Daniel; Zhang, Kang; Goldbaum, Michael (2018), "Labeled Optical Coherence Tomography (OCT) and Chest X-Ray Images for Classification", Mendeley Data, V2, doi: 10.17632/rschjbr9sj.2

### **Random Images Dataset:**

- Kaggle Dataset: Unsplash Random Images Collection
- Used for negative samples (i.e., images that are not chest X-rays).

## **Workflow**

### **1. Image Verification:**

- The API first verifies if the uploaded image is a chest X-ray using the Chest X-ray Detection Model.

### **2. Pneumonia Prediction:**

- If the image is confirmed to be a chest X-ray, it is sent to the Pneumonia Classification Model.
- The model predicts whether pneumonia is present and its type (bacterial or viral).

### **3. Result Display:**

- The prediction results are sent back to the frontend for display.
- Users can then interpret the prediction and act accordingly.

## **Future Directions**

### **Enhanced Model Accuracy:**

Continuous training and fine-tuning of models to improve accuracy and robustness.

### **Expansion of Dataset:**

Incorporating more diverse and extensive datasets to capture a wider range of cases and improve generalization.

## **Run Locally**

### **Client**

This is a Next.js project bootstrapped with `create-next-app`.

First, install all dependencies:

```
npm install  
# or  
yarn  
# or  
pnpm install
```

```
# or  
bun install
```

Then, run the development server:

```
npm run dev  
# or  
yarn dev  
# or  
pnpm dev  
# or  
bun dev
```

Open <http://localhost:3000> with your browser to see the result.

## Server

This is a Python DJANGO project.

First, install all dependencies:

Python 2:

```
pip install -r requirements.txt
```

Python 3:

```
pip3 install -r requirements.txt
```

Then, start the server:

```
python manage.py runserver
```

By default, the server will start and run on <http://localhost:8000>

To save all dependencies in requirements.txt,

```
pip freeze > requirements.txt
```

## About the Author

This project was developed by Tyler Haisman, a computer science student interested in medicine and aspiring to work in technology. For inquiries or collaborations, please contact Tyler via [tylerhaisman.com](http://tylerhaisman.com).

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